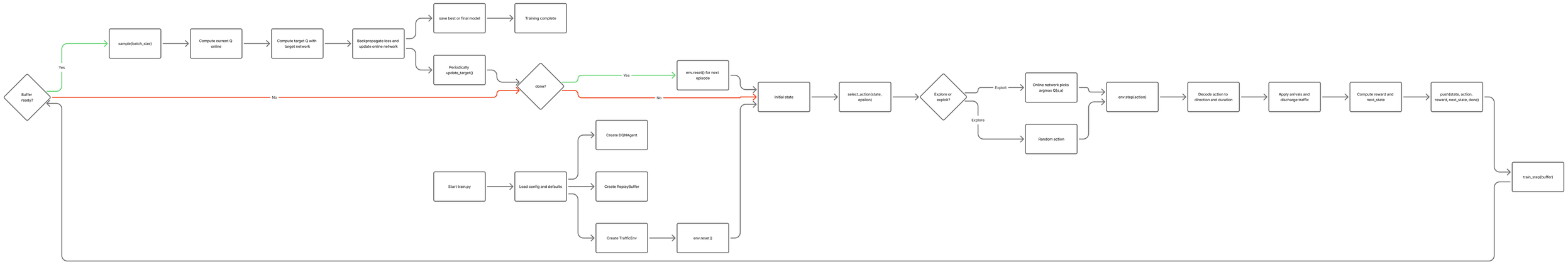
Buffer ready?
Yes
No
sample(batch_size)
Compute current Q online
Compute target Q with target network
Backpropagate loss and update online network
save best or final model
Training complete
Periodically update_target()
done?
Yes
No
env.reset() for next episode
Start train.py
Load config and defaults
Create DQNAgent
Create ReplayBuffer
Create TrafficEnv
env.reset()
Initial state
select_action(state, epsilon)
Explore or exploit?
Exploit
Explore
Online network picks argmax Q(s,a)
Random action
env.step(action)
Decode action to direction and duration
Apply arrivals and discharge traffic
Compute reward and next_state
push(state, action, reward, next_state, done)
train_step(buffer)